



Spatial intelligence refers to the ability to understand, analyze and interpret information related to space

- Acquisition, Analyzing and Interpreting Satellite Imagery, Drone and IoT Imagery
- Utilize technologies such as GIS*, AI**, HPC***
- To understand changes over time to support decision-making in Infrastructure planning, Construction monitoring, Urban planning, Mining operations, Defence and intelligence, Crop monitoring, Forestry and other fields



*

GIS : Geographic information system,

**

AI: Artificial Intelligence,

HPC: High Performance Computing

Image data types

1.

Monochrome

Capture images in shades of grey, useful for tasks like object detection and tracking

2.

Normal

Capture images similar to how the human eye sees them provisioning basic colour and depth information

3.

Stereo Vision

Create 3D information by capturing images from two slightly different viewpoints enabling depth perception and object distance measurement

4.

Multispectral/ Hyperspectral

Capture images across multiple wavelengths, allowing for analysis of material composition and properties

Services we offer

- Geospatial intelligence & remote sensing
- AI for infrastructure monitoring & maintenance
- AI for renewable energy optimization
- Human-centric AI for surveillance & interaction
- Advanced vision & 3D analytics
- Super-resolution & high-definition modeling

Industries we empower

- Infrastructure & Construction
- Energy & Utility
- Defence & Intelligence
- Textile Manufacturers
- Mining & Natural Resources
- Government

Products List

Saptarishi

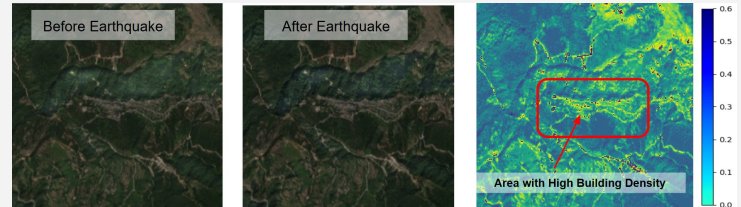
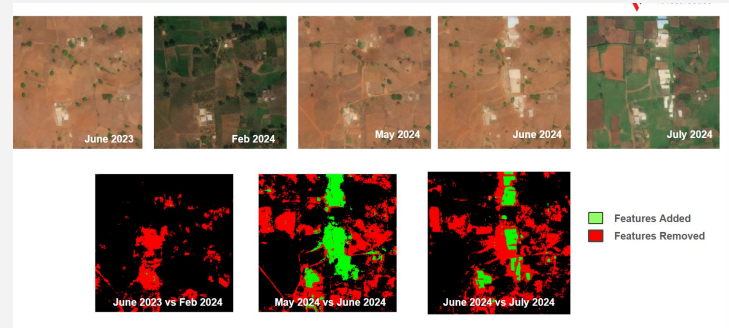
Image agnostic platform for change detection and information retrieval in remote sensing

Key Features

- Multi-modal image analysis (Panchromatic, Multispectral, Hyperspectral, SAR, Weather, Atmospheric).
- **Urban planning:**
 - Detects buildings, structures and urban expansion.
 - Monitors transportation infrastructure changes.
 - Identifies illegal construction & land encroachment.
- **Disaster Management:** Tracks landslides, terrain shifts and embankment changes.
- **Environmental:** Assesses fire damage and burn scars, Deforestation

Technology Stack

- **Imagery:** Aerial, Multispectral, Hyperspectral.
- **AI & ML:** Deep Learning, Reinforcement Learning, Computer Vision.
- **Analysis:** Time-series, Remote Sensing.



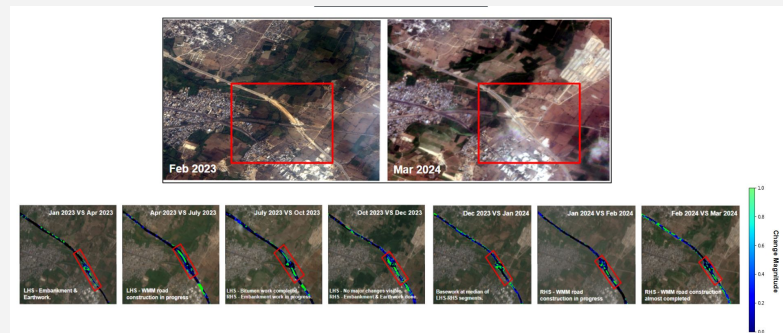
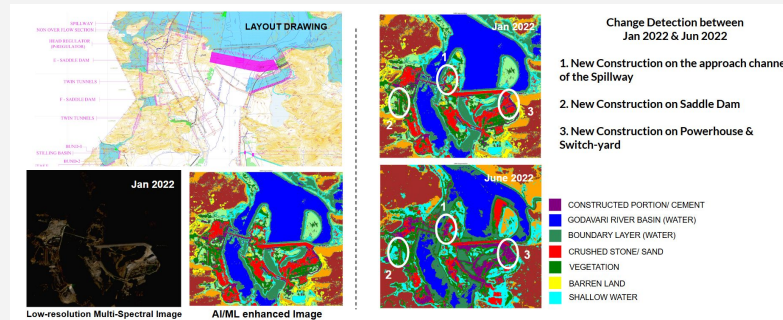
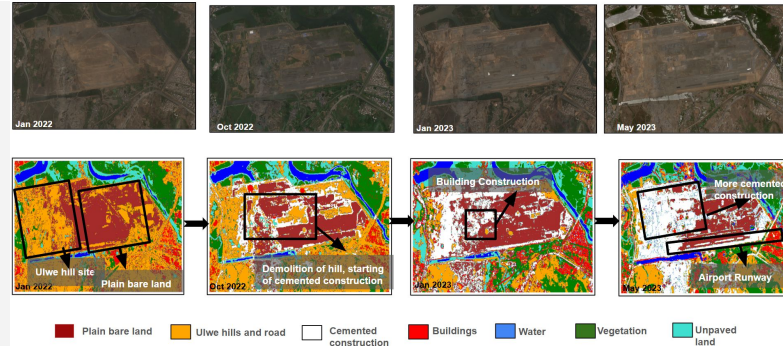
Construction monitoring

Key Features

- **Urban planning:** Tracks construction progress and urban development in real time.
- **Environmental/Agricultural:**
 - Monitors deforestation, reforestation and seasonal vegetation changes.
 - Monitors crop cycles and soil health.
 - Optimizes land use & agricultural planning.

Technology Stack

- **Imagery:** Aerial, Multispectral, Hyperspectral.
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Prahari

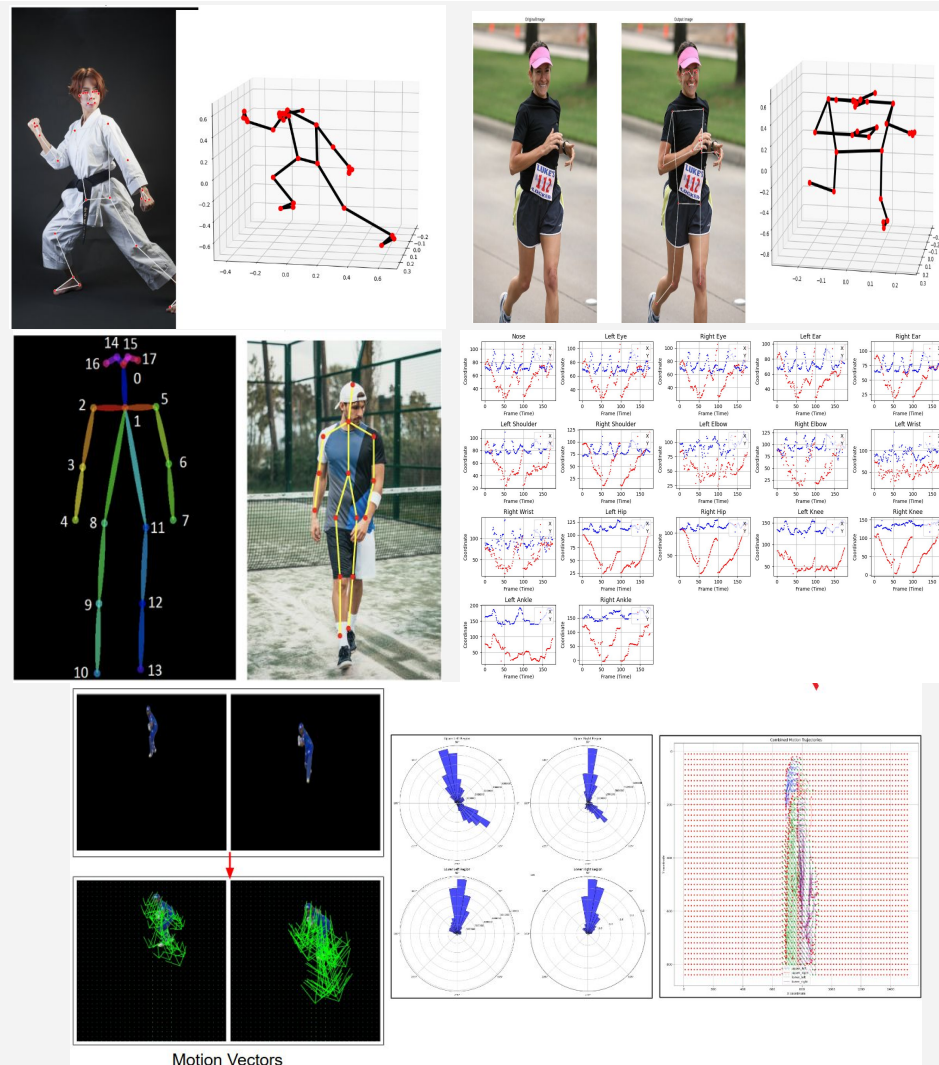
Action Recognition identification and surveillance

Key Features

- Body Keypoints Detection for precise motion analysis.
- Skeleton Representation to model human movement.
- Pose Tracking for real-time activity monitoring.
- Action Similarity to compare and classify movements.
- Multiple applications in sports, healthcare, human-robot interaction and automated surveillance.

Technology Stack

- **AI & ML:** Machine Learning, Deep Learning.
- **Computer Vision:** Keypoint detection & motion analysis.
- **Similarity Metrics:** Gesture comparison & recognition.





Action Recognition

Identification. and surveillance

Sookshma

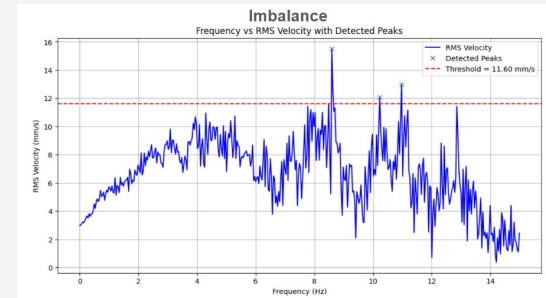
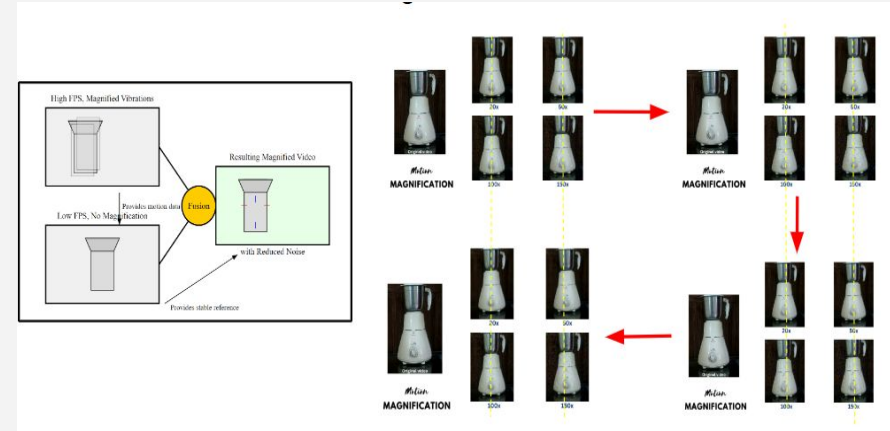
Motion Magnification

Key Features

- Imbalance & Misalignment Detection for early fault identification.
- Mechanical Looseness & Bent Shaft Detection to prevent wear and tear.
- Eccentricity & Resonance Analysis for improved stability.
- Natural Frequency Analysis (Bump Test) to assess structural integrity.
- Electrical Noise Detection for signal accuracy in diagnostics.

Technology Stack

- **Signal Processing:** Fast Fourier Transform, Frequency Filtering.
- **Computer Vision & AI:** Machine Learning, Edge Detection.
- **Vibration Analysis:** Fluid Dynamics Equations, Resonance Models.

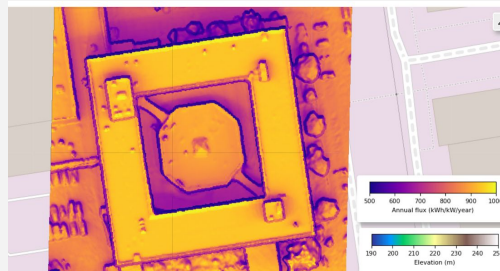
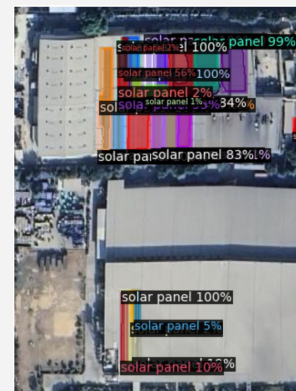
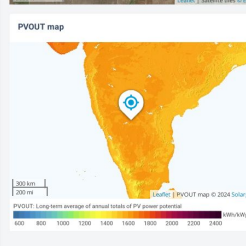
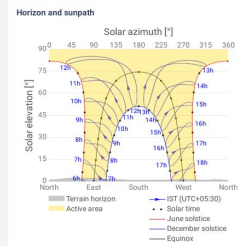


Key Features

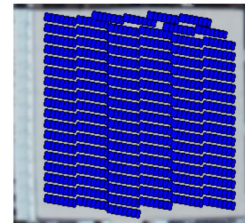
- Solar Panel Detection & Estimation for precise mapping.
- **Solar Irradiance Mapping** (DNI, GHI, DIF, GTI, ELE) for energy potential assessment.
- **Shadow & Shading Analysis** to optimize solar efficiency.
- Climate Science Models for accurate weather impact predictions.
- **Thermal Performance Monitoring** to prevent overheating and efficiency loss.

Technology Stack

- **Imagery:** Aerial, Multispectral, Hyperspectral, Infrared.
- **AI & Computer Vision:** Automated detection & performance analysis.
- **Climate Modeling & Predictive Analytics:** Weather & irradiance forecasting.
- **Remote Sensing:** High-precision solar mapping.



Original Image



Solar Panel Placements



Building height using shadow analysis

ThermoInspect

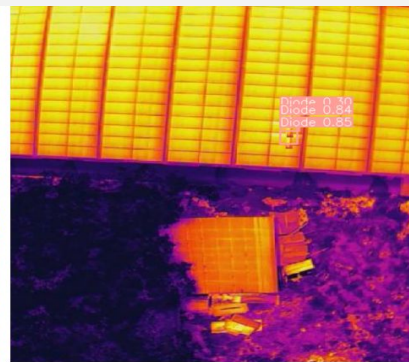
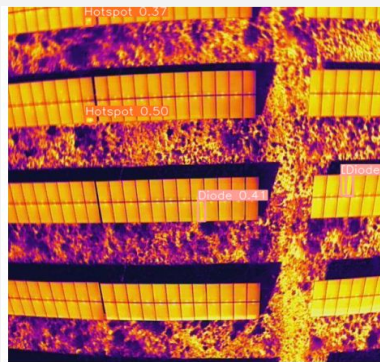
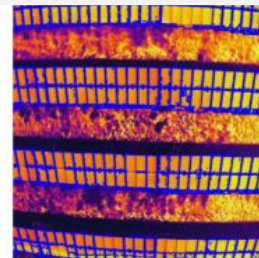
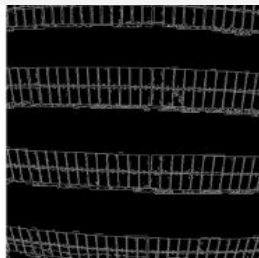
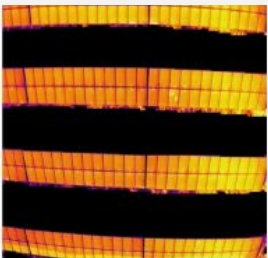
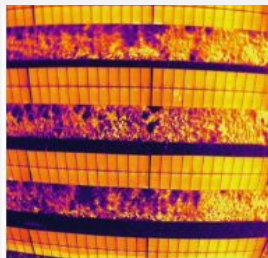
Drone-Based Solar Defect Detection Using Thermal Imaging

Key Features

- Thermal Defect Detection for identifying faulty solar panels.
- Fault Classification to prioritize critical repairs.
- Real-Time Monitoring for continuous system health tracking.
- Drone-Based Inspections for fast, scalable assessments.
- Comprehensive Reporting with actionable insights automated solar farm inspections.

Technology Stack

- **Computer Vision & AI:** Automated fault detection & classification.
- **Deep Learning & Machine Learning:** Predictive maintenance analytics.
- **Thermal Imaging:** Detects hotspots & performance issues.
- **Drone Integration:** Rapid, cost-effective inspections.



TerraVision

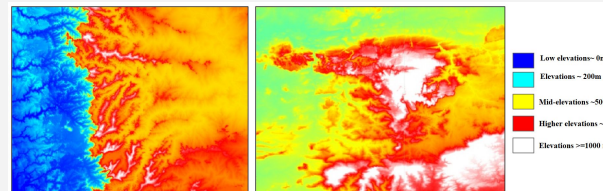
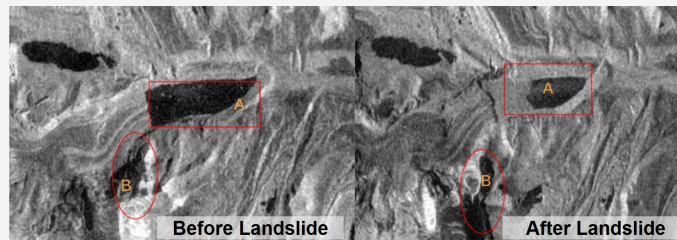
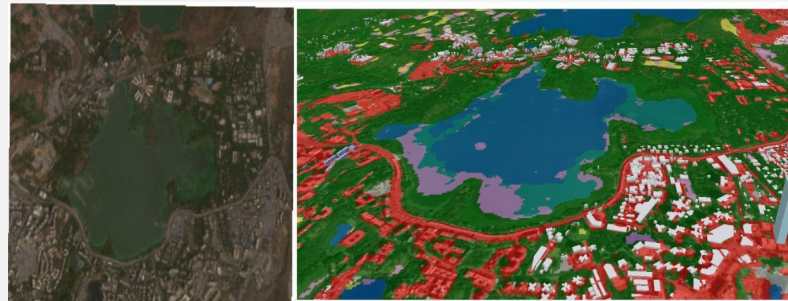
GIS Capabilities

Key Features

- Digital Elevation Model (DEM) for topographic mapping.
- Digital Surface Model (DSM) to analyze surface structures.
- Land Use/Land Cover (LULC) classification for environmental insights.
- SAR Imagery & Analysis for change detection, disaster monitoring, and interferometric analysis.

Technology Stack

- Geospatial Analytics & Remote Sensing for high-precision mapping.
- Aerial & Satellite Imagery for large-scale monitoring.
- Machine Learning & GIS for automated data processing and interpretation.



ResoGen

GenAI

Key Features

- **Super Resolution:** Enhances image clarity and fine details.
- **Super Resolution DEM:** High-precision elevation modeling.
- **Super Resolution DSM:** Improved surface modeling for better geospatial analysis.
- Provides cost-effective and time-saving solutions for urban growth predictions and infrastructure monitoring.

Technology Stack

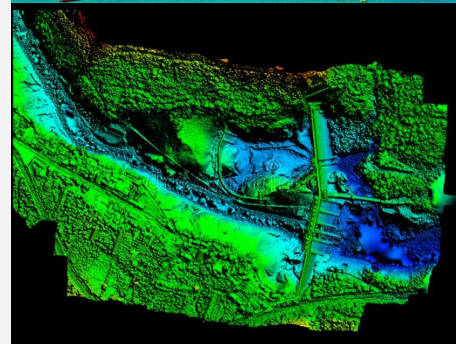
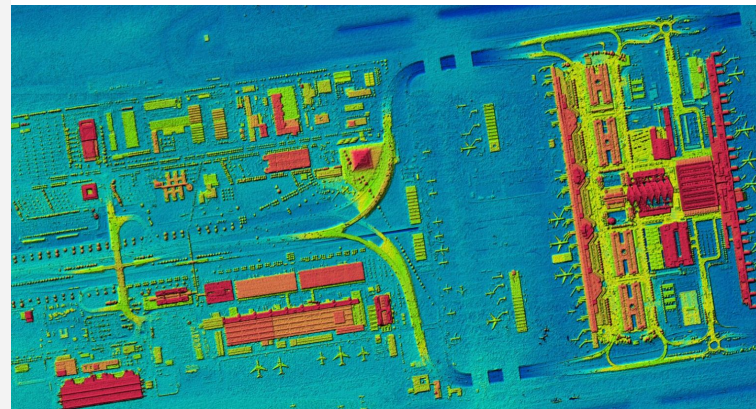
- **Aerial & Satellite Imagery:** High-resolution data acquisition.
- **AI & Machine Learning:** Enhances image resolution and accuracy.
- **Remote Sensing & Image Processing:** Extracts detailed geospatial insights.

Super-Resolution Processing of Low-Resolution Images



Location_1

Location_2



HydroScope

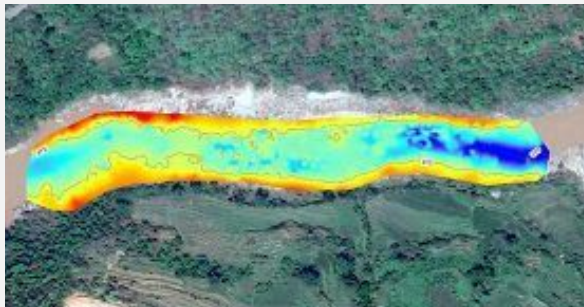
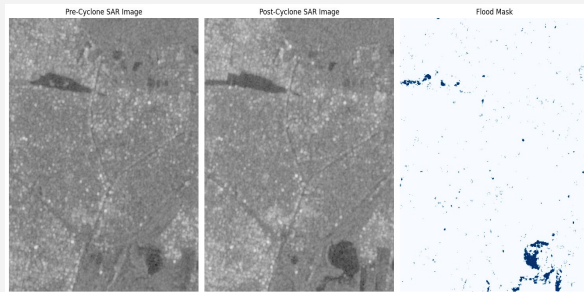
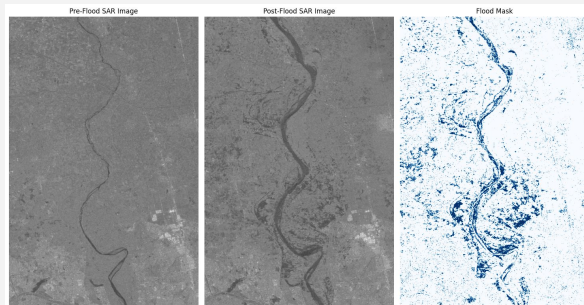
Flood Estimation and monitoring

Key Features

- **Flood Analysis & Masking:** Identifies flood extent and severity.
- **Climate Science Models:** Predicts and assesses climate-related flood risks.
- **Flood Risk Mapping & Impact Assessment:** Evaluates vulnerable areas.
- **Flash Flood Forecasting:** Provides early warnings for disaster mitigation.
- **River Bathymetry:** Analyzes river depth and flow dynamics.
- **Soil Analysis:** Assesses flood impact on agriculture and land stability.

Technology Stack

- **Remote Sensing & GIS:** Captures and processes geospatial data.
- **AI & Machine Learning:** Enhances flood prediction models.
- **Data & Predictive Analytics:** Provides actionable insights.
- **Climate Modeling & Image Processing:** Improves risk assessments and early warnings.



WayVision

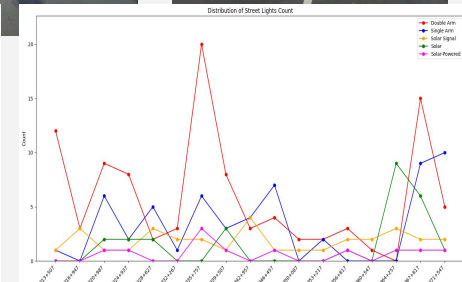
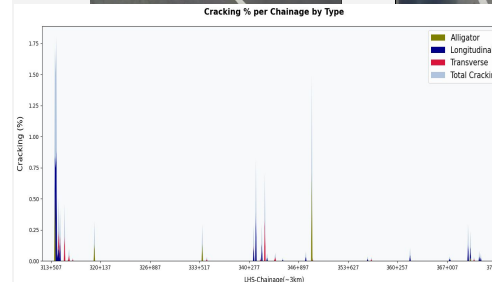
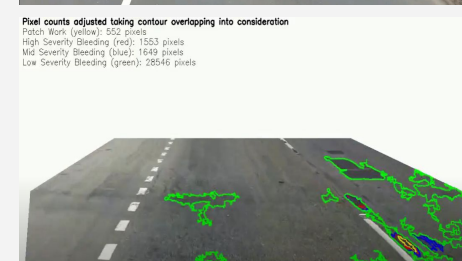
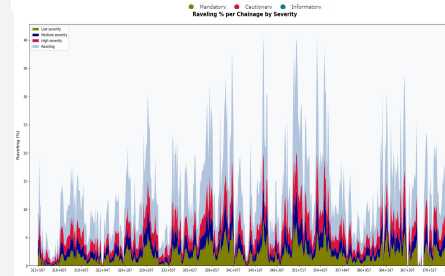
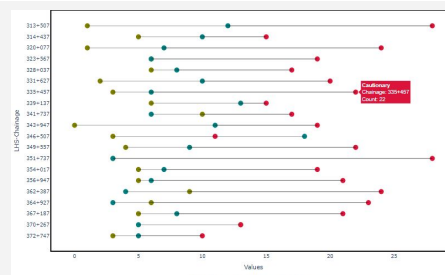
Road object detection & roughness monitoring

Key Features

- **Road Furniture & Street Light Detection:** Identifies and classifies road signs and street light types.
- **Road Defect Detection:** Detects cracks and potholes for maintenance planning.
- **Plant Detection in Medians:** Counts and assesses plant health along road medians.
- **Kerb & Edge Detection**
- **Crash Barrier & Delineator Detection**
- **Road Width & Curvature Measurement:** Evaluates road geometry for planning and safety.
- **Road Roughness Index (RRI):** Assesses road smoothness to determine driving comfort.

Technology Stack

- **Computer Vision:** Processes visual data to detect objects and anomalies.
- **Machine Learning / AI:** Enhances detection accuracy through predictive modeling.
- **Object Detection Algorithms:** Identifies and classifies road-related elements efficiently.



RoadHawk

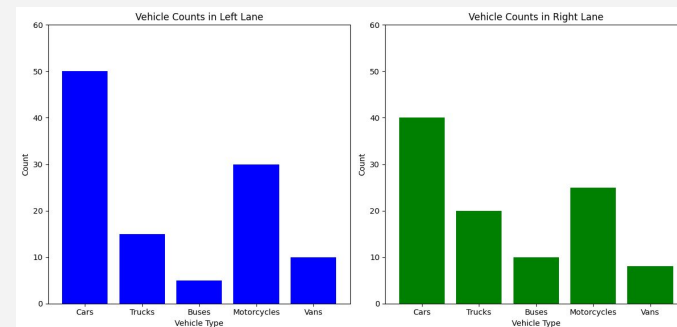
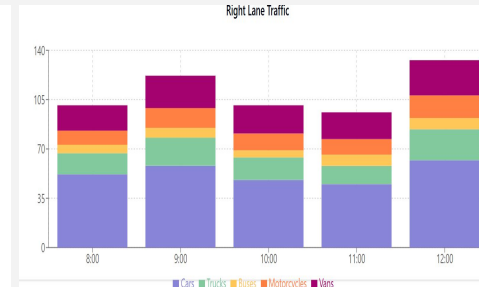
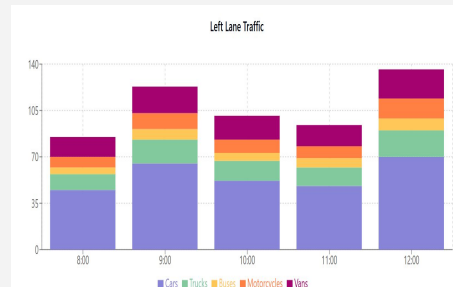
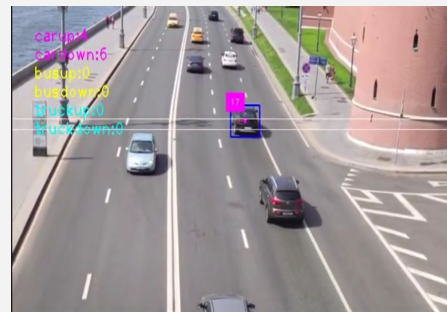
Traffic monitoring

Key Features

- **Vehicle Detection & Counting:** Automatically identifies and tracks vehicles in real-time.
- **Vehicle Classification:** Categorizes vehicles (cars, buses, trucks, motorcycles) for traffic analysis.
- **License Plate Estimation:** Extracts plate numbers for security, toll collection, and law enforcement.
- **Traffic Density Monitoring:** Analyzes congestion levels to optimize road usage and reduce delays.
- Useful in automated toll collection, traffic optimization, parking and access control, security and law enforcement.

Technology Stack

- **Computer Vision:** Enables real-time vehicle detection and classification.
- **Machine Learning / AI:** Enhances accuracy in identifying and tracking vehicles.



BeltRaksha

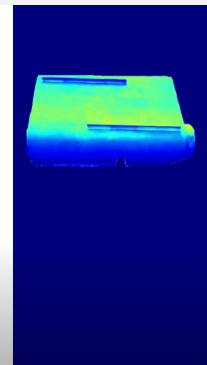
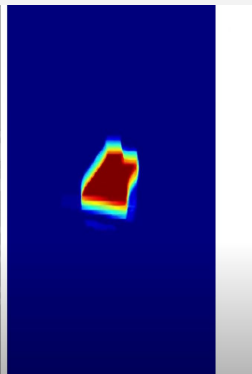
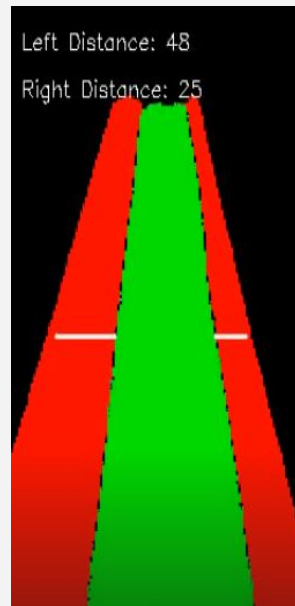
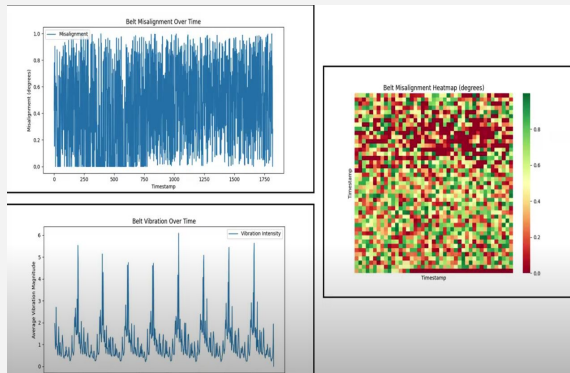
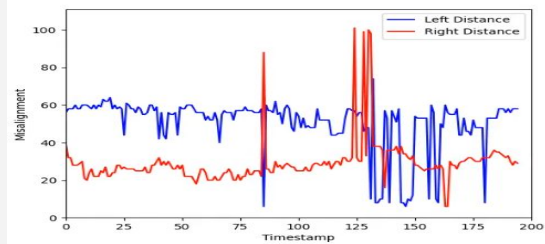
Conveyor Belt Monitoring

Key Features

- **Vibration Analysis:** Detects anomalies in roller and motor performance.
- **Belt alignment & Defect Detection:** Finds misalignment, cracks, wear and other conveyor belt issues.
- **Trend & Root Cause Analysis:** Analyzes historical data to prevent recurring failures.
- **Remaining Useful Life (RUL) Estimation:** Predicts component wear based on visual data.

Technology Stack

- **Computer Vision & Image Processing:** Enables precise defect detection.
- **Machine Learning / AI:** Improves failure prediction and maintenance planning.
- **Predictive Analytics:** Enhances reliability by forecasting potential breakdowns.
- **Vibration Analysis Techniques:** Identifies mechanical imbalances before failure.



StereoSense

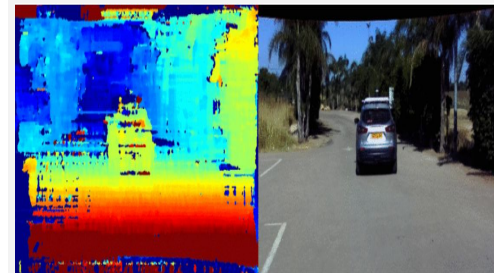
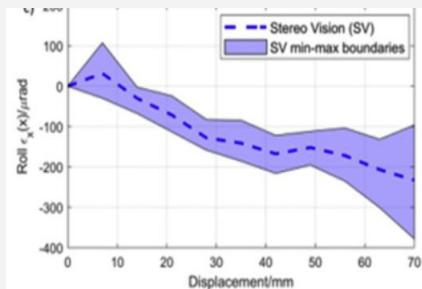
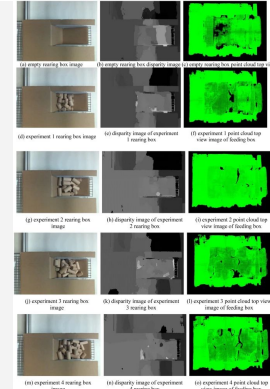
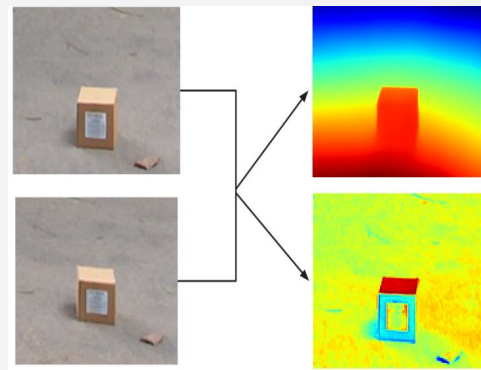
Stereo Vision

Key Features

- **Point Cloud Generation:** Converts image data into 3D point clouds for accurate modeling.
- **Volumetric Analysis:** Measures object volumes for site planning and material estimation.
- **Depth Estimation:** Determines object distances and elevations.
- **Surface Area Calculation:** Computes surface dimensions for precise analysis.
- **Object Detection & Measurement:** Identifies and quantifies objects in 3D space.

Technology Stack

- **Computer Vision & Stereo Imaging:** Captures depth and spatial relationships.
- **3D Reconstruction:** Builds detailed models from multi-angle data.
- **Depth Sensing:** Enhances object recognition and navigation.
- **Machine Learning & 3D Rendering:** Improves accuracy in object classification and visualization.



SnapUniform

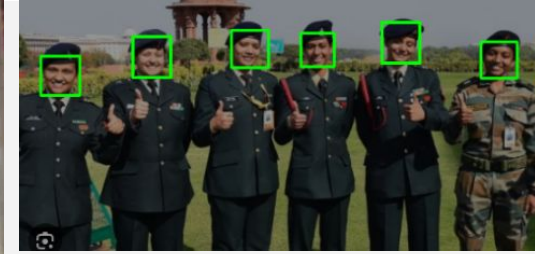
Face and Uniform Detection System

Key Features

- **Facial Recognition:** Identifies individuals for security and authentication.
- **Multi-Face Detection:** Detects and tracks multiple faces in real time.
- **Uniform Detection:** Recognizes specific uniforms for workforce and security personnel monitoring.
- **Multi-Object Detection:** Identifies and classifies various objects in an environment.
- Useful in crowd monitoring, workforce tracking, retail analytics and healthcare.

Technology Stack

- **Computer Vision & Deep Learning:** Enhances facial and object recognition accuracy.
- **Facial Landmark Detection:** Identifies key facial features for precise recognition.
- **Object Detection Algorithms:** Detects and classifies multiple objects efficiently.
- **Pattern Recognition:** Analyzes features in images for improved identification.



Solara

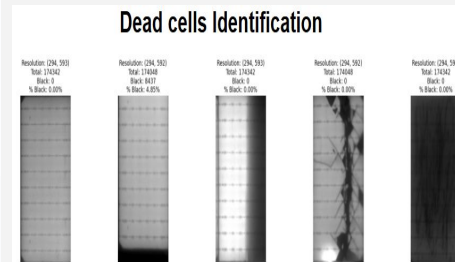
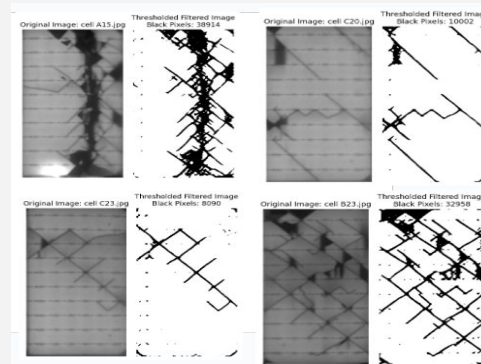
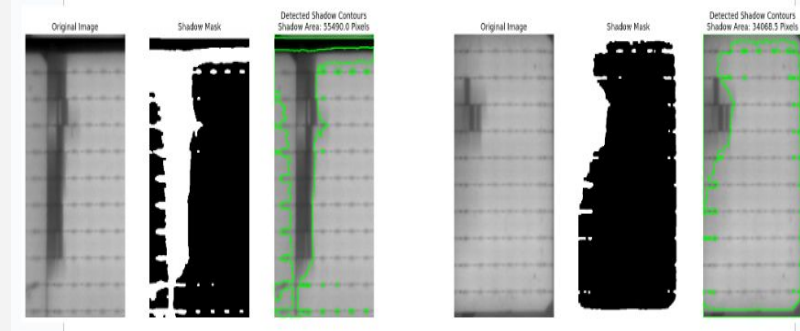
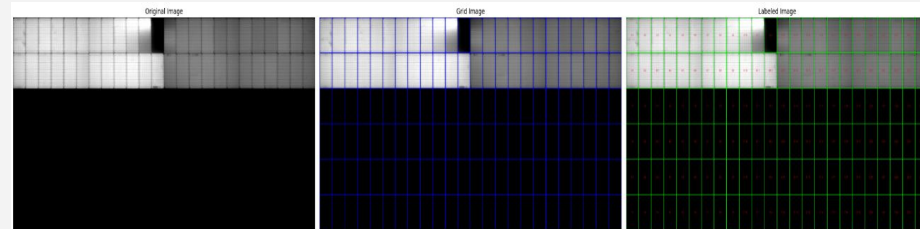
Solar panel shadow and crack analysis

Key Features

- **Cell Segmentation:** Identifies individual solar cells for precise defect detection.
- **Defect Localization & Pixel Value Estimation:** Pinpoints defects with pixel-level accuracy.
- **Severity Analysis of Defects:** Classifies damage levels to prioritize maintenance.
- **Dead Cell Detection:** Identifies non-functional cells affecting energy efficiency.

Technology Stack

- **Computer Vision & Image Processing:** Enables detailed defect analysis.
- **Machine Learning / AI:** Improves accuracy in defect classification and severity assessment.

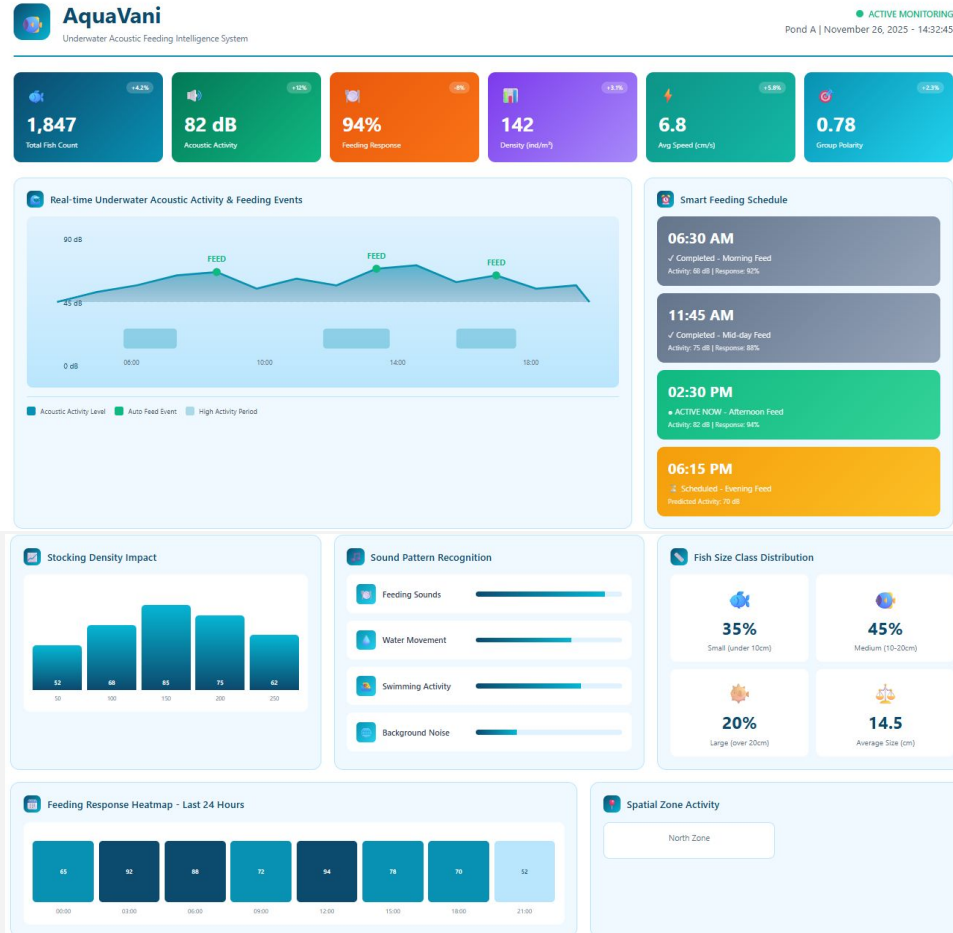


Key Features

- Real-time monitoring of fish feeding sounds to detect feeding intensity and behavior patterns.
- Automated feeding control to reduce feed wastage and improve aquaculture efficiency.
- AI-based anomaly detection for identifying stress, overfeeding, or unhealthy fish activity.
- Data analytics dashboard for tracking feeding trends, fish growth, and farm performance.

Technology Stack

- **Audio Processing:** FFT, spectrogram analysis, noise filtering, acoustic signal processing.
- **AI/ML Models:** Deep learning and pattern recognition models for fish behavior classification.
- **Hardware & Sensors:** Underwater microphones (hydrophones), IoT devices, edge computing units.



Construction Eye

AI powered construction monitoring site

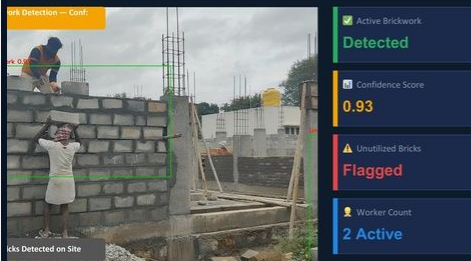
Key Features

- Real-Time AI Vision Monitoring — Continuously analyzes live video streams to deliver instant, automated site intelligence around the clock
- Multi-Module Detection System — Covers construction activity, heavy equipment, safety compliance, and material inventory within a single integrated platform
- Automated Safety Compliance Alerts — Instantly identifies and flags safety violations without any human review.
- Digital Twin Visualization — Translates real-world site conditions into an interactive spatial dashboard
- Material Quantification & Wastage Control — Automatically counts and tracks on-site resources to minimize losses and optimize supply planning
- Scalable Multi-Camera Architecture — Deployable across multiple camera feeds simultaneously.
- Status Classification Engine — Goes beyond detection to classify operational states, distinguishing between active, idle, and completed conditions in real time

Technology Stack

- Computer Vision , Instance Segmentation ,Object Tracking ,Digital Twin Engine

RUCTION MONITORING



AI-POWERED CONSTRUCTION MONITORING

SAFETY MONITORING — UNSAFE WORKER DETECTION



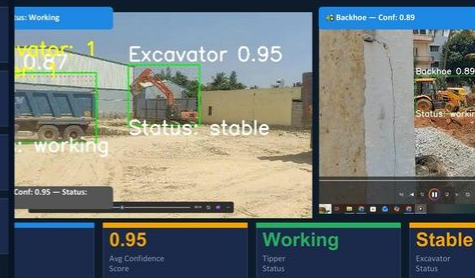
AI-POWERED CONSTRUCTION MONITORING

MATERIAL MONITORING — CEMENT BAGS & SAND



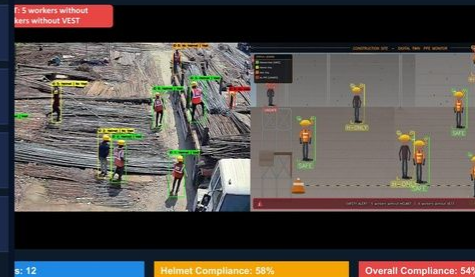
AI-POWERED CONSTRUCTION MONITORING

EQUIPMENT MONITORING — EXCAVATOR & TIPPER



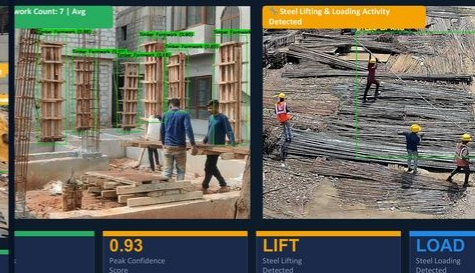
AI-POWERED CONSTRUCTION MONITORING

COMPLIANCE DASHBOARD — DIGITAL TWIN VIEW



AI-POWERED CONSTRUCTION MONITORING

MATERIAL MONITORING — TIMBER FORMWORK & STEEL OPERATIONS



AI-POWERED CONSTRUCTION MONITORING

CONSTRUCTION MONITORING

MODULE 01



Active Brickwork
Detected

Confidence Score
0.93

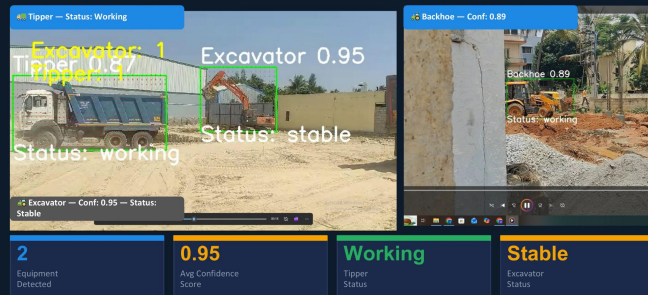
Unutilized Bricks
Flagged

Worker Count
2 Active

SPATIALTY.AI | AI-POWERED CONSTRUCTION MONITORING

EQUIPMENT MONITORING — EXCAVATOR & TIPPER

MODULE 02



SPATIALTY.AI | AI-POWERED CONSTRUCTION MONITORING

MATERIAL MONITORING — CEMENT BAGS & SAND



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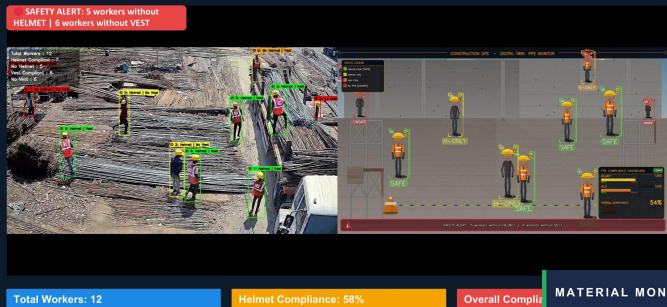
PPE SAFETY MONITORING — UNSAFE WORKER DETECTION

MODULE 03



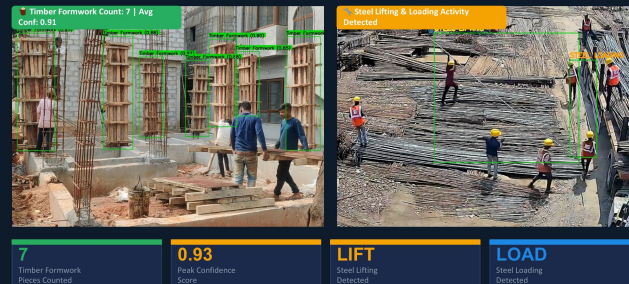
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PPE COMPLIANCE DASHBOARD — DIGITAL TWIN VIEW



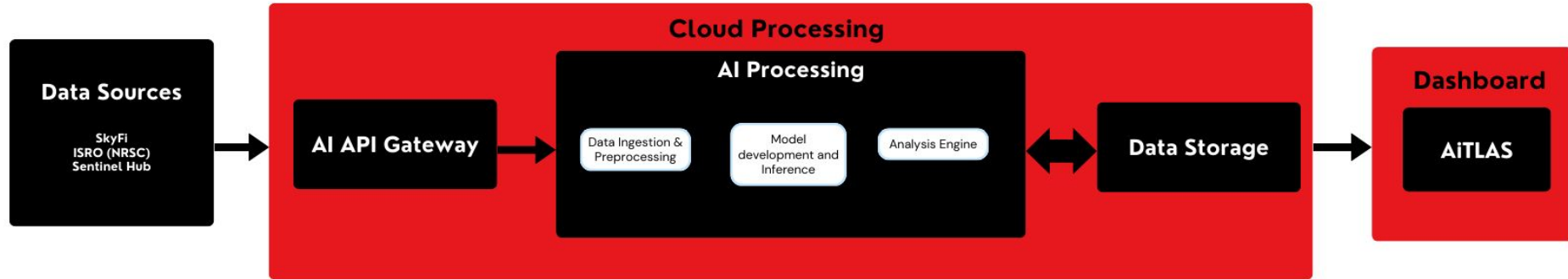
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MATERIAL MONITORING — TIMBER FORMWORK & STEEL OPERATIONS



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AI as a Service (AlaaS)



Our Clientele (Non-exhaustive)

Frischmann Prabhu

FP Project Management
Formerly Frischmann Prabhu (India) Private Limited



Angelique International Ltd.



Western M P Infrastructure & Tolls
Private Limited (WMP)

N.A.M. EXPRESSWAY LTD

Our Partners

Digital PMC & Risk Consulting Partners

CRISIL

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*PS: MoU with **CRISIL Consulting - Transport, Logistics & Mobility (TLM) group** to offer Digital PMO and Risk Consulting aided by AI-based Geospatial Intelligence Services to Infrastructure and Capital Projects clients*



A Venture of Tagya Research Private Limited

Registered Address:

*#1, ADDEVISHWANATHAPURA, ARKARE CHOKHANHALLI ROAD,
RAJANKUNTE, BANGALORE- 560064*

Phone: +91 8618398926

Email: contact@spatialty.ai

CIN: U72200KA2022PTC169451
